



2.6

Competing Function Model Validation

Student Notes and Guided Practice

AP Precalculus - Unit 2

| ECHS Mathematics

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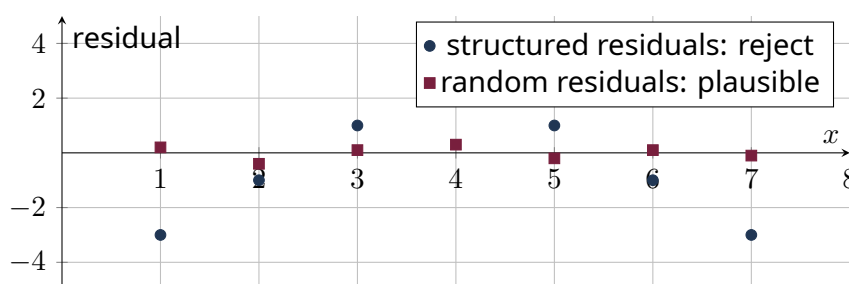
Ghuwaleh

Learning Targets

- Compare candidate models using residuals and contextual structure.
- Quantify signed, absolute, and relative errors.
- Explain consequences of overprediction and underprediction.

Essential Question

How do residual patterns and context help validate or reject competing models?



Concept Framework

Key Idea

Residual sign reveals direction: positive means observed exceeds predicted.

Key Idea

Random residual scatter supports a model; curves or trends indicate missed structure.

Key Idea

Small errors alone are not enough if assumptions or domain are inappropriate.

Key Idea

Consequences of error depend on context, not only magnitude.

Formula Map**Formula and Structure**

$$e = y - \hat{y}$$

Formula and Structure

$$\text{MAE} = \frac{1}{n} \sum |e_i|$$

Formula and Structure

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum e_i^2}$$

Worked Examples**Worked Example 1**

Problem. Observed value is 72 and predicted value is 76. Find the residual.

Solution. $72 - 76 = -4$; the model overpredicted by 4.

Worked Example 2

Problem. Why can RMSE be more sensitive to outliers than MAE?

Solution. RMSE squares errors, giving large residuals disproportionately greater weight.

Worked Example 3

Problem. A model has tiny in-sample error but implausible long-run behavior. Is it validated?

Solution. Not necessarily. Structural plausibility and intended domain are also required.

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Observed=50, predicted=47. Compute and interpret residual; include relative error when meaningful.

2. Observed=80, predicted=86. Compute and interpret residual; include relative error when meaningful.

3. Observed=12, predicted=10. Compute and interpret residual; include relative error when meaningful.

4. Observed=200, predicted=190. Compute and interpret residual; include relative error when meaningful.

5. Observed=0, predicted=3. Compute and interpret residual; include relative error when meaningful.

6. Model A residuals: -2,1,0,1,-2. Model B: -4,0,0,0,4. Compare MAE and RMSE qualitatively.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.

Video support: [Flipped Math topic page](#) | Original EHS material informed by Pearson and Holt Precalculus concepts.